

BME 313L: Introduction to Numerical Methods in Biomedical EngineeringHonor Code: *I have neither given nor received aid on this quiz.*

UT EID: _____

Printed Name: _____

Signature: _____

Problem 1 (10 Points): A respiratory gating system tracks a tumor that moves along a straight line as the patient breathes. The tumor position is modeled by

$$p(t) = \frac{\sin(2t)}{2} + \frac{2}{5}$$

The beam must switch on when the tumor first reaches a position of 0.5. We will use Newton's method for root finding to determine t^* such that $p(t^*) = 0.5$. The code below implements Newton's method to determine the beam-on time, but some of it is missing. Fill in the missing code.

```

01 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
02 function t = beamTime
03 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
04 % Initial guess
05 t = 0.2;
06
07 % Maximum iterations
08 maxIter = 50;
09
10 % Stopping criteria
11 tol = 1e-8;
12
13 % Newton iteration
14 for i = 1:maxIter
15
16     % Function evaluation
17     [f,df] = tPos(t);
18
19     % Newton step
20     tNew = ????????
21
22     % Compute the approximate relative error
23     err = ?????;
24
25     % Update t
26     t = tNew;
27
28     % Check convergence
29     if( err < tol )
30         ????????
31     end
32
33 end
34
35 % Compute p(t) and its derivative
36 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
37 function [p,dp] = tPos(t)
38 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
39 p = sin(2*t)/2 + 2/5;
40 dp = ????????

```

Line 20: `tNew = t - (f - 0.5)/df;`

Line 23: `err = abs(tNew - t)/abs(tNew);`

Line 30: `break`

Line 40: `dp = cos(2*t);`

Problem 2 (9 pts): A second patient's gating problem provides only a black-box routine for the residual $g(t)$, with no derivative available. We wish to find t^* such that $g(t^*) = 0$. Evaluating it at $t_l = 0.0$ and $t_u = 1.2$ gives $g(t_l) = -0.2$ and $g(t_u) = 0.6$. **(A)** What is the bisection method's approximation to the root after the first iteration?

ANSWER A) (3 pts): 0.6

Bisection takes the midpoint of the bracket and ignores the function values entirely:

$$(0.0 + 1.2)/2 = 0.6$$

B) Using the same bracket and the same two function values, compute the first iterate produced by the method of false position and explain in one sentence why it differs from your answer to part A).

ANSWER B) (6 pts): 0.3

$$\begin{aligned} t_r &= t_u - g(t_u)(t_l - t_u) / (g(t_l) - g(t_u)) \\ &= 1.2 - (0.6)(0.0 - 1.2) / (-0.2 - 0.6) \\ &= 1.2 - 0.9 = 0.3 \end{aligned}$$

False position replaces g by the straight line through the two endpoints and takes the root of that line. A useful check: the iterate must lie inside the bracket

Problem 3 (6 Points): Two iterative methods are applied to the same root finding problem. The true absolute error $|x_k - x^*|$ is recorded on four consecutive iterations of each:

Method A: 1×10^{-1} , 2×10^{-3} , 8×10^{-7} , 1.28×10^{-13}

Method B: 1×10^{-1} , 2.5×10^{-2} , 6.25×10^{-3} , 1.5625×10^{-3}

A) Classify the convergence of each method as linear or quadratic, and justify your answer.

ANSWER A) (4 pts): Method A is quadratic, Method B is linear.

Method A, ratio of each error to the square of the one before it:

$$2e-3/(1e-1)^2 = 0.2 \quad 8e-7/(2e-3)^2 = 0.2 \quad 1.28e-13/(8e-7)^2 = 0.2$$

The ratio is constant, which is the definition of quadratic convergence, with $M = 0.2$.

Method B, ratio of consecutive errors:

$$2.5e-2/1e-1 = 0.25 \quad 6.25e-3/2.5e-2 = 0.25 \quad 1.5625e-3/6.25e-3 = 0.25$$

Constant, so the error is multiplied by a fixed factor each iteration. That is linear.

B) For the method you classified as linear, estimate the constant c in $|x_{k+1} - x^*| \approx c|x_k - x^*|$.

ANSWER B) (2 pts): $c = 0.25$

Every consecutive ratio for Method B is exactly 0.25, so the error is quartered each iteration.

Problem 4 (5 Points): The quadratic $f(x) = x^2 - x - 6$ has roots $x^* = 3$ and $x^* = -2$. Two possible rearrangements of $f(x) = 0$ into the form $x = g(x)$ are

$$g_1(x) = \sqrt{x+6} \quad \text{and} \quad g_2(x) = x^2 - 6$$

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BME 313L Quiz 3 - ANSWERS

Which formulation should be used to find the root with fixed point iteration? Justify your answer. Hint: Compute $g_1'(x)$ and $g_2'(x)$, and evaluate each at $x^* = 3$.

ANSWER (5 pts): Use g_1 .

$$\begin{aligned} g_1'(x) &= 1/(2*\sqrt{x+6}) \\ g_2'(x) &= 2x \end{aligned}$$

$$\begin{aligned} g_1'(3) &= 1/6 \\ g_2'(3) &= 6 \end{aligned}$$

Fixed-point iteration converges to x^* when $|g'|$ stays below 1 on an interval containing x^* . At $x^* = 3$ the first rearrangement gives $1/6$, comfortably inside the condition, so the iteration contracts and converges. The second gives 6, so any starting point other than 3 exactly is pushed away and the iteration diverges.

Both are algebraically valid rearrangements of the same equation and both have 3 as a fixed point, so the derivative test is the only way to tell them apart.